ridge regressor only produces single-valued predictions, the GP outputs a normal distribution at each test point. Uncertainty quantification is what sets the GP apart from other machine learning models, specifically in the context of decision-making under uncertainty.

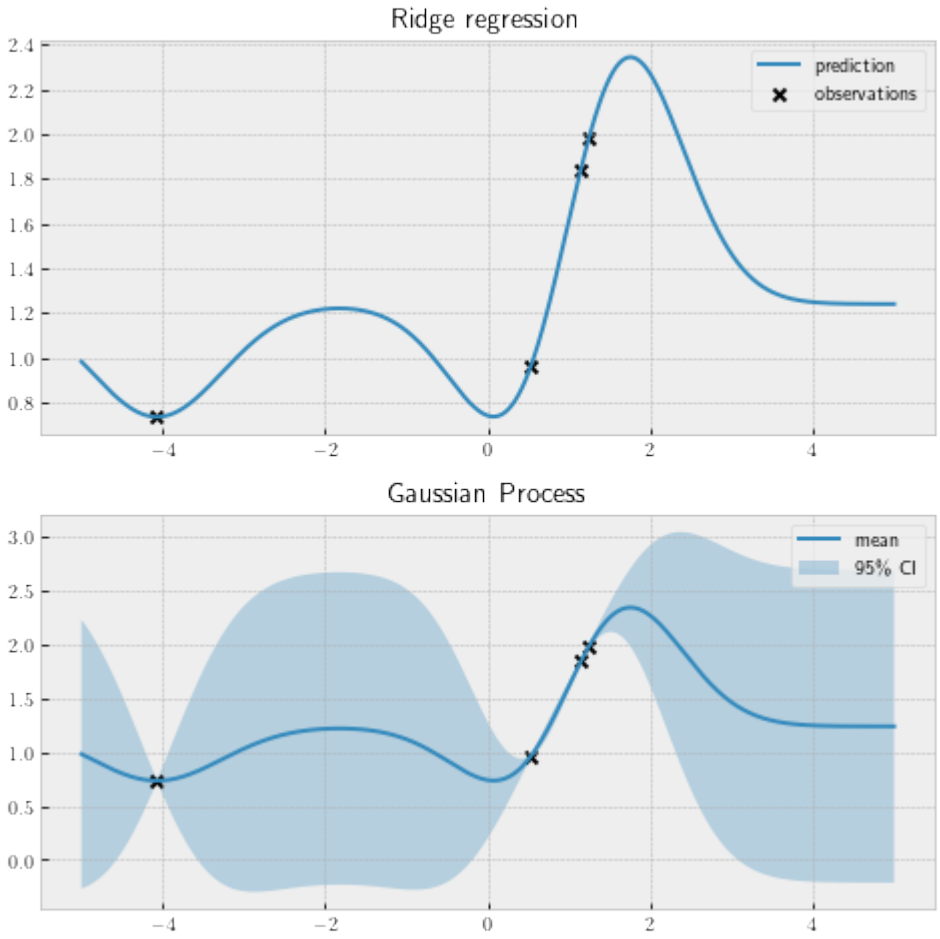


Figure 2.1 Predictions by ridge regression vs. those by a Gaussian process (GP). While the mean prediction of the GP is the same as the prediction of ridge, the GP also offers credible intervals indicating the predictive level of uncertainty.

We will see how correlation modeling and uncertainty quantification are mathematically realized with Gaussian distributions, and learn to actually implement a GP in GPyTorch, the premiere GP modeling tool in Python. Being able to model a function with a GP is the first step toward Bayesian optimization, a step that we will take in this chapter.

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